

Hyunseo Shin

Contact : (+82)010-4938-6248, **Email :** hseo98@naver.com

RESEARCH INTEREST

I am interested in developing reliable language models by improving their alignment, safety, and interaction with external knowledge.

EDUCATION

University of Seoul Mar 2024 – Feb 2026
M.S. in Artificial Intelligence

University of Seoul Mar 2020 – Feb 2024
B.S. in Environmental Engineering, Minor in Statistics

WORK EXPERIENCE

Kakaobank Sep 2026 – *Present*
AI Research Engineer (Contract)

University of Seoul Mar 2026 – May 2026
Graduate Researcher

NAVER Place AI Sep 2025 – Oct 2025
Machine Learning Engineer (Intern)
Project : Developed a place-domain-specific embedder and constructed retrieval train/test datasets.

PUBLICATIONS

(*:1st author)

1) Seungwon Woo*, Hyunseo Shin*, Wonseok Hwang. "In-Context Learning via Attention Patching." 2026. (*submitted*)

2) Kyeongjun Cho, Juheon Kang, Hyunseo Shin, Hongseok Oh, Wonseok Hwang. "FTC: Fourier Transcoder for Monosemantic Sparse Feature Decomposition in Vision Language Models." ACCV 2026.

3) Juheon Kang, Eunkyung Choi, Hongseok Oh, Hyunseo Shin, JuYoun Son, Yunjin Bae, Wonseok Hwang. "Q-CAT: A Non-Parametric Query Filtering System for Prohibited Content in Dynamic Enterprise Environments." EMNLP Main 2026.

4) Hyunseo Shin*, Gio Paik*. "Towards Truly Multilingual ASR: Generalizing Code-Switching ASR to

Unseen Language Pairs." *ICML Workshop on Machine Learning for Audio*, 2026.

5) Hyunseo Shin*, Wonseok Hwang. "Layer-wise Swapping for Generalizable Multilingual Safety." *EACL Main* 2026.

6) Hyunseo Shin*, Youngrok Choi, Wonseok Hwang. "Causal Retrieval via Semantic Regularization." *IEEE Access* 2026.

7) Seungwon Woo*, Hyunseo Shin*, Eunkyung Choi*, Juheon Kang*, and Wonseok Hwang. "K-SNAC: Robust Neuron Coverage for OOD Generalization and Test Adequacy." *IEEE ASE Workshop on Explainable Automated Software Engineering*, 2025.

8) Hyunseo Shin* and Wonseok Hwang. "Causal Retrieval with Semantic Consideration." *NeurIPS 2024 Workshop on Causal Representation Learning*, 2024.

9) Yeeun Kim*, Hyunseo Shin, Eunkyung Choi, Hongseok Oh, Hyunjun Kim, and Wonseok Hwang. "On the Consideration of AI Openness: Can Good Intent Be Abused?" *AAAI Conference on Artificial Intelligence*, 2025.

PROJECT

[산학과제] 삼성 **SDS B2B** 사업 환경에서의 **LLM** 안전성 향상 연구 Oct 2025 – May 2026

[국가과제] **Meta AI Alignment for AI Safety** Sep 2025 – Feb 2026

안전한 AI를 위한 메타 AI 정렬 기술

- 다국어·저자원 조건에서 모델의 안정성 및 Alignment 성능 개선 연구

[국가과제] 기질 특이적 **AI-기반 고활성 신규 효소공학기술 개발** Jun 2024 – Feb 2026

ESM-3 를 활용한 단백질 구조-기질 결합 특성 반응을 위한 모델 개발

- 단백질의 3차원 구조 및 서열 정보를 바탕으로, 효소-기질 간 결합 상호작용을 예측 모델 구축

[교내 과제] 교내 규정 검색 서비스 개발 Jun 2023 – Dec 2023

Team member, Project during Undergraduate Internship

- GPT 및 ROUGE 기반 평가를 통한 직원 업무 Retrieval Dataset 구축

Music For Your Day Sep 2022 – Jan 2023

Team Leader, Project in the club TAVE

감성 분석을 기반으로 음악 추천 시스템

- 감정 기반 음악 추천 모델 구현 및 Gradio 기반 사용자 웹페이지 제작

AWARDS

대학원생 연구역량 강화 프로그램 Jan 2026

서울시립대학교 일반대학원, 우수상

‘저자원 언어에서 안정성을 강화하기 위한 레이어별 스와핑 기반 학습 전략’
연구

전문지식 활용 학술심화 및 사회문제해결 자율·융합 연구과제

Dec 2024

서울시립대학교 일반대학원, 우수상

‘다중흡 환경에서 인과관계를 고려한 검색’

CERTIFICATES

빅데이터분석기사	2023.07.14
SQLD	2022.09.30
ADsP	2022.09.23